

LCM3 Instruction set (alphabetic order)

Syntax	Operation	Values	Code																	
			15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
ADD Rd,Rn,#imm3	$Rd \leftarrow Rn + ZE(imm3)$	$0 \leq imm3 \leq 7$	0	0	0	1	1	1	0	imm3			n			d				
ADD Rd,#imm8	$Rd \leftarrow Rd + ZE(imm8)$	$0 \leq imm8 \leq 255$	0	0	1	1	0	d			imm8 (unsigned)									
ADD Rd,Rn,Rm	$Rd \leftarrow Rn + Rm$		0	0	0	1	1	0	0	m			n			d				
AND Rd,Rm	$Rd \leftarrow Rd \text{ AND } Rm$		0	1	0	0	0	0	0	0	0	0	m			d				
B label	$PC \leftarrow \text{adr}(\text{label})$	$-2^{10} \leq imm11 \leq 2^{10}-1$	1	1	1	0	0	imm11 (Δi in 2's complt)												
BXX label	if Z=0 then $PC \leftarrow \text{adr}(\text{label})$	$-2^7 \leq imm8 \leq 2^7-1$	1	1	0	1	XX=cond			imm8 (Δi in 2's cmpt)										
CMP Rn,#imm8	$NZVC \leftarrow Rn - ZE(imm8)$	$0 \leq imm8 \leq 2^8-1$	0	0	1	0	1	n			imm8 (unsigned)									
EOR Rd,Rm	$Rd \leftarrow Rd \oplus Rm$		0	1	0	0	0	0	0	0	0	1	m			d				
LDR Rt,[Rn]	$Rt \leftarrow M[\text{align4}(Rn)]$		0	1	1	0	1	0	0	0	0	0	n			t				
LSL Rd,Rm,#imm5	$Rd \leftarrow Rm \ll imm5$	$0 \leq imm5 \leq 31$	0	0	0	0	0	imm5			m			d						
MOV Rd,#imm8	$Rd \leftarrow ZE(imm8)$	$0 \leq imm8 \leq 255$	0	0	1	0	0	d			imm8 (unsigned)									
MOV Rd,Rm	$Rd \leftarrow Rm$		0	0	0	0	0	0	0	0	0	0	m			d				
STR Rt,[Rn]	$M[\text{align4}(Rn)] \leftarrow Rt$		0	1	1	0	0	0	0	0	0	0	n			t				
SUB Rd,Rn,#imm3	$Rd \leftarrow Rn - ZE(imm3)$	$0 \leq imm3 \leq 7$	0	0	0	1	1	1	1	imm3			n			d				
SUB Rd,#imm8	$Rd \leftarrow Rd - ZE(imm8)$	$0 \leq imm8 \leq 255$	0	0	1	1	1	d			imm8 (unsigned)									
SUB Rd,Rn,Rm	$Rd \leftarrow Rn - Rm$		0	0	0	1	1	0	1	m			n			d				

In "BXX label", XX : is a condition (if the condition is true then branch to label)

XX=NE $\leftrightarrow$ Z=0 (Not Equal) $\leftrightarrow$ cond=0001

XX=EQ $\leftrightarrow$ Z=1 (is Equal) $\leftrightarrow$ cond=0000

XX=GE $\leftrightarrow$ N=V (Greater or Equal) $\leftrightarrow$ cond=1010

XX=LT $\leftrightarrow$ N !=V (Lower Than) $\leftrightarrow$ cond=1011

XX=GT $\leftrightarrow$ Z=0 et N=V (Greater Than) $\leftrightarrow$ cond=1100

XX=LE $\leftrightarrow$ Z=1 ou N !=V (Less or equal) $\leftrightarrow$ cond=1101

Not implemented yet in LCM3

ZE: Zero Extend; extension of the value to 32 bits (unsigned)

The variables n, d, m or imm3 are unsigned.

Only one instruction modifies the flags NZVC here: it's « CMP ».

Mw[adr] is a word (32 bits) in memory at address adr.

Remark : adr must be a multiple of 4: 0, 4, 8, 0xC, 0x10...

The function align4(i)= (i/4)*4 : it's equivalent to clear the 2 LSBs.

LCM3 Instruction set (binary code order)

Syntax	Operation		Code																		
			15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0			
LSL Rd, Rm, #imm5	$Rd \Leftarrow Rm \ll imm5$	$0 \leq imm5 \leq 31$	0	0	0	0	0	imm5				m				d					
ADD Rd, Rn, Rm	$Rd \Leftarrow Rn + Rm$		0	0	0	1	1	0	0	m				n				d			
SUB Rd, Rn, Rm	$Rd \Leftarrow Rn - Rm$		0	0	0	1	1	0	1	m				n				d			
ADD Rd, Rn, #imm3	$Rd \Leftarrow Rn + ZE(imm3)$	$0 \leq imm3 \leq 7$	0	0	0	1	1	1	0	imm3				n				d			
SUB Rd, Rn, #imm3	$Rd \Leftarrow Rn - ZE(imm3)$	$0 \leq imm3 \leq 7$	0	0	0	1	1	1	1	imm3				n				d			
MOV Rd, #imm8	$Rd \Leftarrow ZE(imm8)$	$0 \leq imm8 \leq 255$	0	0	1	0	0	d				imm8 (unsigned)									
CMP Rn, #imm8	$NZVC \Leftarrow Rn - ZE(imm8)$		0	0	1	0	1	n				imm8 (unsigned)									
ADD Rd, #imm8	$Rd \Leftarrow Rd + ZE(imm8)$	$0 \leq imm8 \leq 255$	0	0	1	1	0	d				imm8 (unsigned)									
SUB Rd, #imm8	$Rd \Leftarrow Rd - ZE(imm8)$	$0 \leq imm8 \leq 255$	0	0	1	1	1	d				imm8 (unsigned)									
AND Rd, Rm	$Rd \Leftarrow Rd \text{ AND } Rm$		0	1	0	0	0	0	0	0	0	0	m				d				
EOR Rd, Rm	$Rd \Leftarrow Rd \oplus Rm$		0	1	0	0	0	0	0	0	0	1	m				d				
STR Rt, [Rn]	$M[\text{align4}(Rn)] \Leftarrow Rt$		0	1	1	0	0	0	0	0	0	0	n				t				
LDR Rt, [Rn]	$Rt \Leftarrow M[\text{align4}(Rn)]$		0	1	1	0	1	0	0	0	0	0	n				t				
BXX label	if Z=1 then $PC \Leftarrow \text{adr}(\text{label})$	$-2^7 \leq imm8 \leq 2^7 - 1$	1	1	0	1	cond				imm8 ($\Delta i 2's c$)										
B label	$PC \Leftarrow \text{adr}(\text{label})$	$-2^{10} \leq imm11 \leq 2^{10} - 1$	1	1	1	0	0	imm11 ($\Delta i 2's c$)													



LCM3: why?

- The LCM3 instruction set is compatible with the Cortex-M3 instruction set. It's in fact a part of the Cortex-M3 instruction set.
- The LCM3 instructions are easy to understand and not numerous. We can't say that for Cortex-M3 instructions.
- Beginning with the Cortex-M3 instructions is rather difficult: you must use a tool that is not as friendly as LCM3 simulator.
- You'll be able to program a real Cortex-M3 microcontroller by using LCM3 instructions.
- The LCM3 is RISC architecture like Cortex-M3 but is easier to understand.

LCM3 Simulator has the following features:

➤ EDITOR

- Assembler mnemonics are highlight as they are recognized (whoa!).
- Save and load code.
- The rules to write your code:
 - One mnemonic for one line
 - A mnemonic must begin after a space or a tabulation
 - If you define a label, its first character must be in the first column

➤ ASSEMBLER

- Active when the 'Assembly' « assemblage » button is pressed.
- Assembler key words are checked.
- Operands and immediate value range are checked.
- If the code syntax is correct, it downloads it in ROM memory from 0x08000000
- 3 instructions and a label are added at the end of your code: "FIN B FIN" "MOVE R0,#0" and "MOVE R0,R0".

➤ SIMULATOR

- The « pas à pas » button is here to execute step by step your instructions.
- « Reset »: it resets memory and registers contents to 0.
- Onglet ROM : visualisation de la ROM : adresse, code et instruction assembleur
- « CodeBinaire » tab: is here to visualize the binary code of the mnemonics.
- Registers, RAM, flags and pipeline are visualized.